Instalación y configuración de Instancias Vagrant

HashiCorp

Vagrant

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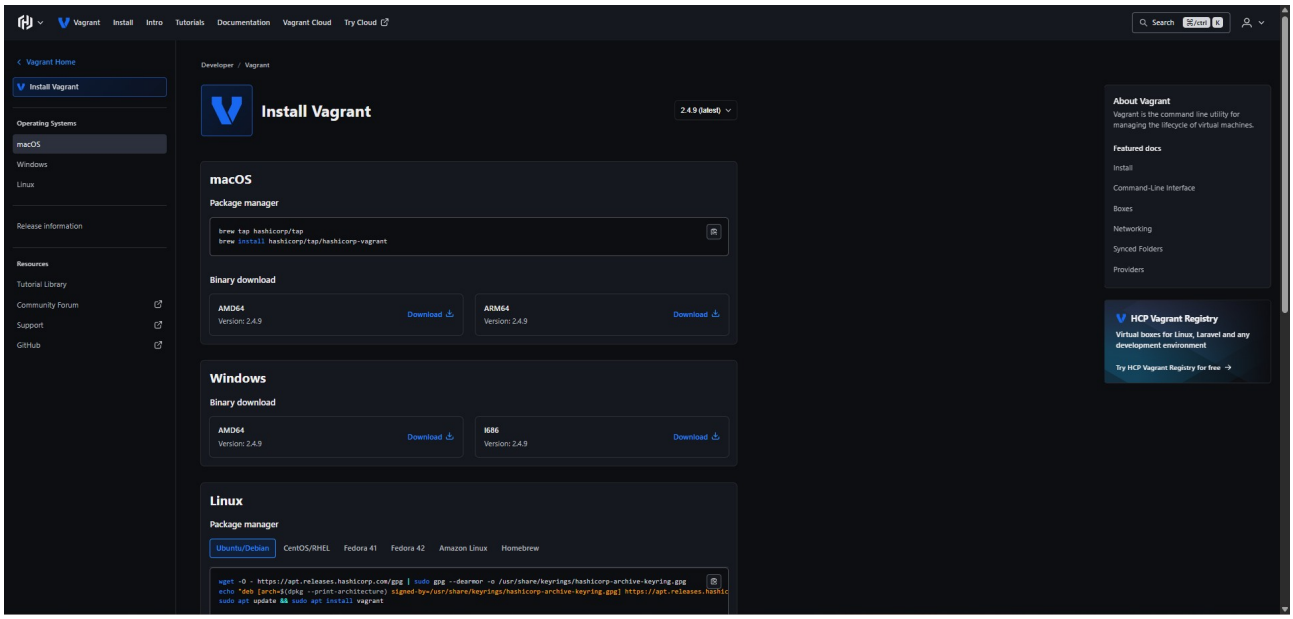
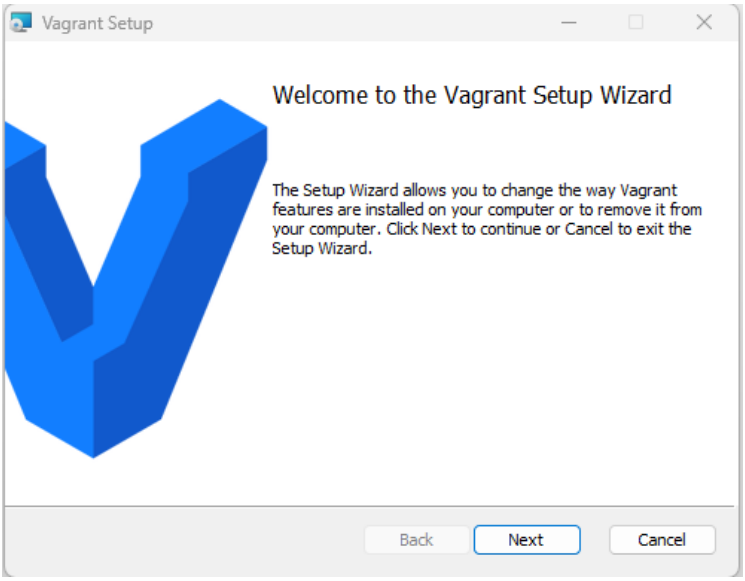
Descargamos Vagrant

Lo primero que haremos sera entrar a la pagina oficial de Vagrant y descargarnoslo
https://developer.hashicorp.com/vagrant



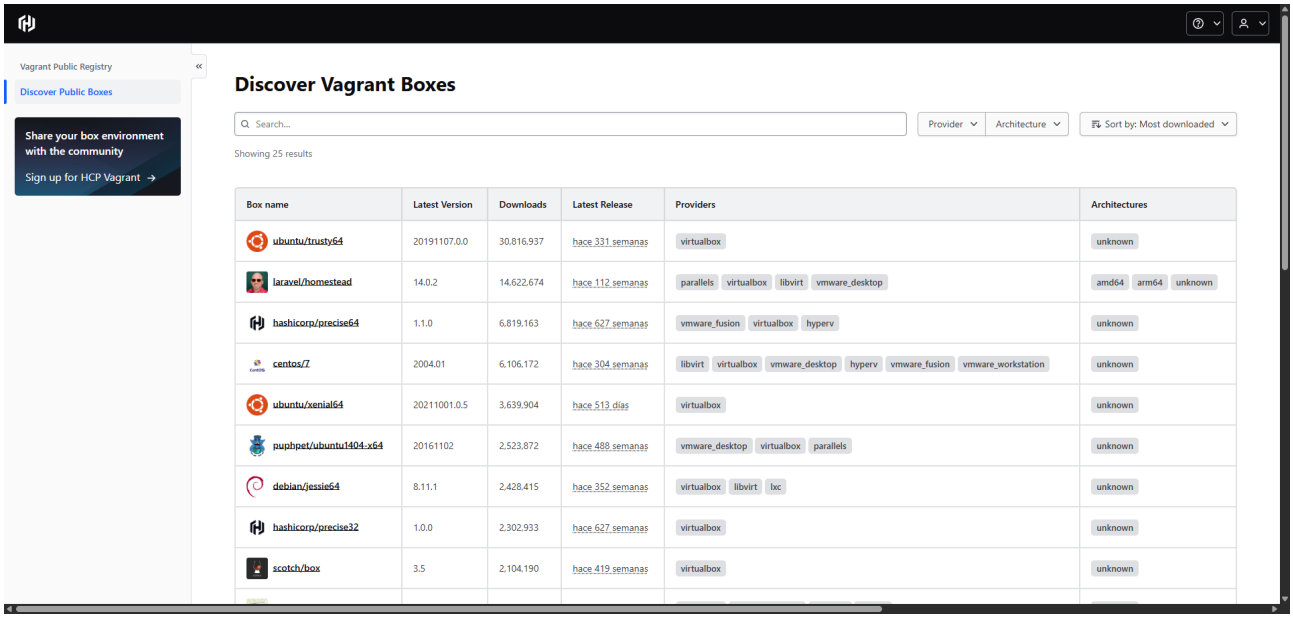
Vemos que esta varias versiones para varios sistemas operativos acompañado de la explicación para la instalación si es necesaria



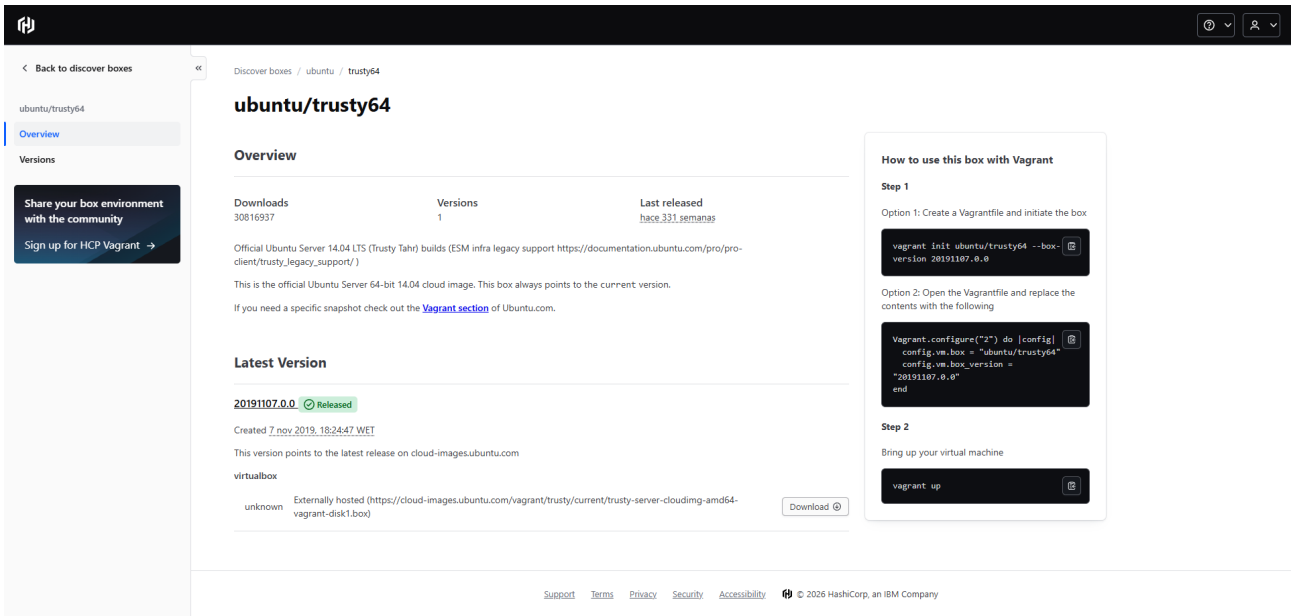
Una vez descargado, abrimos el instalador y terminamos de instalar Vagrant

Una vez este instalado todo ya estará listo, sin embargo explicare que tambien hay una pagina para distintas versiones de vagrant y como usarlas

https://portal.cloud.hashicorp.com/vagrant/discover



Esta es la interfaz con todas las posibles versiones que puedes utilizar



Y dentro de cualquier versión, hay su forma de instalarlo

Configuración de Vagrant

```
Símbolo del sistema
Microsoft Windows [Versión 10.0.26200.8037]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\guill>vagrant init
A `Vagrantfile` has been placed in this directory. You are now
ready to `vagrant up` your first virtual environment! Please read
the comments in the Vagrantfile as well as documentation on
`vagrantup.com` for more information on using Vagrant.

C:\Users\guill>
```

Una vez este todo instalado, el primer paso para crear nuestras instancias es abrir la CMD y escribir el comando ***"vagrant init"***, una vez hecho eso, ya se nos crearía el Vagrantfile que es donde ira toda la configuración de las instancias

```
Vagrantfile
C: > Users > guill > Vagrantfile
# -*- mode: ruby -*-
# vi: set ft=ruby :

# All Vagrant configuration is done below. The "2" in Vagrant.configure
# configures the configuration version (we support older styles for
# backwards compatibility). Please don't change it unless you know what
# you're doing.
Vagrant.configure("2") do |config|
  # The most common configuration options are documented and commented below.
  # For a complete reference, please see the online documentation at
  # https://docs.vagrantup.com.

  # Every Vagrant development environment requires a box. You can search for
  # boxes at https://vagrantcloud.com/search.
  config.vm.box = "base"

  # Disable automatic box update checking. If you disable this, then
  # boxes will only be checked for updates when the user runs
  # `vagrant box outdated`. This is not recommended.
  # config.vm.box_check_update = false

  # Create a forwarded port mapping which allows access to a specific port
  # within the machine from a port on the host machine. In the example below,
  # accessing "localhost:8080" will access port 80 on the guest machine.
  # NOTE: This will enable public access to the opened port
  # config.vm.network "forwarded_port", guest: 80, host: 8080

  # Create a forwarded port mapping which allows access to a specific port
  # within the machine from a port on the host machine and only allow access
  # via 127.0.0.1 to disable public access
  # config.vm.network "forwarded_port", guest: 80, host: 8080, host_ip: "127.0.0.1"

  # Create a private network, which allows host-only access to the machine
  # using a specific IP.
  # config.vm.network "private_network", ip: "192.168.33.10"

  # Create a public network, which generally matched to bridged network.
  # Bridged networks make the machine appear as another physical device on
  # your network.
  # config.vm.network "public_network"

  # Share an additional folder to the guest VM. The first argument is
  # the path on the host to the actual folder. The second argument is
  # the path on the guest to mount the folder. And the optional third
  # argument is a set of non-required options.
  # config.vm.synced_folder "../data", "/vagrant_data"

  # Disable the default share of the current code directory. Doing this
```

Este es el vagrantfile que se nos genera por defecto, ahora nosotros le daremos nuestra forma

Lanzamiento de las instancias

```
C:\Users\jasos\TFG>vagrant up
Bringing machine 'borg-server' up with 'virtualbox' provider...
Bringing machine 'borg-client' up with 'virtualbox' provider...
==> borg-server: Checking if box 'ubuntu/jammy64' version '20241002.0.0' is up to date...
==> borg-server: Clearing any previously set forwarded ports...
==> borg-server: Clearing any previously set network interfaces...
==> borg-server: Specific bridge 'Realtek PCIe GbE Family Controller' not found. You may be asked to specify
==> borg-server: which network to bridge to.
==> borg-server: Preparing network interfaces based on configuration...
    borg-server: Adapter 1: nat
    borg-server: Adapter 2: bridged
==> borg-server: Forwarding ports...
    borg-server: 22 (guest) => 2222 (host) (adapter 1)
==> borg-server: Configuring storage mediums...
==> borg-server: Running 'pre-boot' VM customizations...
==> borg-server: Booting VM...
==> borg-server: Waiting for machine to boot. This may take a few minutes...
    borg-server: SSH address: 127.0.0.1:2222
    borg-server: SSH username: vagrant
    borg-server: SSH auth method: private key
==> borg-server: Machine booted and ready!
[borg-server] GuestAdditions 7.1.16 running --- OK.
==> borg-server: Checking for guest additions in VM...
==> borg-server: Setting hostname...
==> borg-server: Configuring and enabling network interfaces...
==> borg-server: Mounting shared folders...
    borg-server: C:/Users/jasos/TFG => /vagrant
==> borg-server: Machine already provisioned. Run `vagrant provision` or use the `--provision`
==> borg-server: flag to force provisioning. Provisioners marked to run always will still run.
==> borg-client: Checking if box 'ubuntu/jammy64' version '20241002.0.0' is up to date...
==> borg-client: Clearing any previously set forwarded ports...
==> borg-client: Fixed port collision for 22 => 2222. Now on port 2200.
==> borg-client: Clearing any previously set network interfaces...
==> borg-client: Specific bridge 'Realtek PCIe GbE Family Controller' not found. You may be asked to specify
==> borg-client: which network to bridge to.
==> borg-client: Preparing network interfaces based on configuration...
    borg-client: Adapter 1: nat
    borg-client: Adapter 2: bridged
==> borg-client: Forwarding ports...
    borg-client: 22 (guest) => 2200 (host) (adapter 1)
==> borg-client: Configuring storage mediums...
==> borg-client: Running 'pre-boot' VM customizations...
==> borg-client: Booting VM...
==> borg-client: Waiting for machine to boot. This may take a few minutes...
    borg-client: SSH address: 127.0.0.1:2200
    borg-client: SSH username: vagrant
    borg-client: SSH auth method: private key
    borg-client: Warning: Connection reset. Retrying...
    borg-client: Warning: Connection aborted. Retrying...
==> borg-client: Machine booted and ready!
[borg-client] GuestAdditions 7.1.16 running --- OK.
==> borg-client: Checking for guest additions in VM...
==> borg-client: Setting hostname...
==> borg-client: Configuring and enabling network interfaces...
==> borg-client: Mounting shared folders...
    borg-client: C:/Users/jasos/TFG => /vagrant
==> borg-client: Machine already provisioned. Run `vagrant provision` or use the `--provision`
==> borg-client: flag to force provisioning. Provisioners marked to run always will still run.

C:\Users\jasos\TFG>
```

Con el comando "***vagrant up***" iniciamos las instancias y vemos que se han cargado correctamente

Vemos como en el cliente dentro del virtualbox se ve como están iniciadas las dos instancias y se ve dentro de virtualbox por que es el sistema de virtualizacion que hemos elegido

```
C:\Users\jasos\TFG>vagrant ssh borg-server
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-164-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Mar 17 17:23:41 UTC 2026

  System load:             0.0
  Usage of /:              6.9% of 38.70GB
  Memory usage:            9%
  Swap usage:              0%
  Processes:               101
  Users logged in:         0
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:28:41ff:fee4:55a4

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

39 updates can be applied immediately.
33 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


Last login: Thu Mar 12 19:50:56 2026 from 10.0.2.2
vagrant@borg-server:~$
```

```
C:\Users\jasos\TFG>vagrant ssh borg-client
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-171-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Mar 17 17:24:13 UTC 2026

  System load:             0.0
  Usage of /:              8.6% of 38.70GB
  Memory usage:            10%
  Swap usage:              0%
  Processes:               102
  Users logged in:         0
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:28:41ff:fee4:55a4

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

6 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


Last login: Sun Mar 15 12:35:19 2026 from 10.0.2.2
vagrant@borg-client:~$
```

Una vez esten iniciadas y todo vaya correcto, para acceder a la maquina necesitarías por el comando "***vagrant ssh***" junto al nombre de la maquina

Y LISTO